BIOGRAPHICAL SKETCH

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NAME: Smotherman, Nicholas

eRA COMMONS USER NAME (credential, e.g., agency login): nicksmotherman1

POSITION TITLE: PhD Student (with little Idea what he's doing)

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
Texas A&M University, College Station	BS	08/2018	12/2022	Biological and Agricultural Engineering
University of Central Florida	MS	08/2025	05/2027	Biomedical Engineering
University of Central Florida	PHD	08/2025	05/2030`	Biomedical Engineering

A. Personal Statement

My research is focused generally on the proteomics of neural cells and tissue engineering. I have a diverse background working with many departments, with lab training in biological engineering, mechanical engineering and veterinary medicine. I have worked under many experienced PI's and post-doctoral workers on experiments funded by either the NSF &/or NIH on projects centered around genetics, neuroinflammation, and somatosensory structures. I have handled data analysis and neural tissue harvesting for an NSF graduate fellow working in veterinary medicine. I have designed experimental setups working in tandem with many collaborators on tight deadlines. Consequently, I have learned how critically important it is to actively and always communicate with collaborators, to narrow the scope of a gigantic project to something realistically achievable and verifiable, and to prioritize that which stands to benefit society the most. It is my firm belief that I hold the motivation, personality, and expertise required to carry out this proposal.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2025 – Present Graduate Teaching Assistant, Mechanical and Aerospace Department, University of Central Florida
2022 – 2025 Laboratory Technician, Messina Hof Winery
2020 – 2022 Undergraduate Researcher

Honors

2022 Poster Presenter, TAMU Student Research Week - Somatosensory Pilot Experiment
2022 2nd Place Poster in Science, Engineering, Technology, and Computer Science, TAMU Student Research Week

C. Contributions to Science

Undergraduate Research: My first year of experience with undergraduate research was in the lab of Dr. Candice B.L. at the department of veterinary integrative biosciences at Texas A&M University. The project was run by 3 PhD students with Dr. Brinkmeyer-Langford as the PI. The goal of the project was to investigate how genetic background influences neurodegenerative disease progression using the collaborative cross mouse model. Mice were infected with Theiler's virus to create a mouse model of neuroinflammation. To properly track how genetics played a role, multiple generations of mice had to live a normal life span while being examined twice a day. This necessitated a considerable amount of individual evaluation, which was accomplished using undergraduate assistant researchers, like me. We all participated in the daily neurodegenerative evaluation of the mice, which took place in a secure BSL3 laboratory. Typical for a large quantity of unpaid undergrads, there was a high turnover rate. As such, I quickly found myself as one of the few assistants that had been there longer than a week or two. After a month I was occasionally asked to train a new hire in the evaluation procedure. Leading up to that first training session, I was sweating and nervous. I was afraid I would manage to forget something critical or instruct them wrong somehow. While it was indeed very stressful at first, after a few minutes I was able to get into rhythm, and it was quite fun! Before then, I had never really had the opportunity to train anyone in any real scientific process. Apart from the mouse evaluations, a few times a month those of us that had lasted a while were invited to assist with euthanasia and tissue harvesting, which I found surprisingly interesting. This assistant position provided me with knowledge on the level of effort, organization, and time needed to perform a large-scale experiment that generates a massive quantity of data. I also received a considerable amount of training in handling laboratory animals in a secure environment, as well as research compliance and biosafety protocols. I finished my time at this lab at the end of my sophomore year. I worked in this lab for one year before the P.I. shifted focus away from this project.

Next, in 2020 I joined a joint Mechanical & Biomedical Engineering Laboratory called the INVENT Lab under Dr. M. Cynthia Hipwell, with Dr. Yuan Ma, a post-doctoral researcher who acted as my project manager. For my first project, I was paired up with another student to develop a way to quantify the effectiveness of a haptic wave-focusing technique to be used in telemedicine. Our goal was to build an array of actuators that would fit on a theoretical virtual reality haptic glove, exerting pressure whenever the user interacted with something virtually. One of the largest problems with small-scale actuator arrays was that they do not output a large enough force individually to be useful. The hypothesis was that we could use wave-focusing, very similar to cell phone towers, to generate a wave that would feel more impactful despite using weak actuators. My work was building a complimentary sensor array that would validate the efficacy of our setup. After considerable literature review, I proposed both a 3D PDMS-based sensor grid and an appropriate parallel sampling data acquisition system that would be able to detect if we could target the actuators as anticipated. Shortly after we presented the final design, my partner graduated, Dr. Ma took a job at Hong Kong Polytechnic, and I was reassigned to a new project. This was my first time being entrusted with the independent development of an experimental project. My post-doc advisor was there to check up on me and make sure I had what I needed, but no one really "held my hand" through this process. I believe the self-motivated experience I gained here was invaluable to my growth towards being an independent researcher. I also believe high-volume signal sampling, and working with human-flesh analogues will benefit me greatly in neural tissue engineering in the future.

My second year at INVENT, I was assigned to a team tasked with developing a pilot experiment for discovering virtual haptic dissonance thresholds. When people are in virtual reality, there is a level of spatial inconsistency with their own body and with the real world that our brain is willing to ignore. Various psychological studies have shown that if the eyes are provided with information about a person's surroundings that is inconsistent with what their other senses report, it heavily reduces the viability of a virtual environment. This led to the conclusion that at some quantifiable threshold of incongruence, there is a disjuncture where the person's mind rejects the illusion in front of their eyes. Our experiment was to evaluate this phenomenon when it came to touch. We subdivided our pilot into two parts, using active touch and passive touch. Passive touch is especially important to virtual reality, as haptic gloves would provide similar feedback to a stationary human hand. We provided human subjects with simple polygons for them to feel with their hands and graded their ability to identify the shape while having their hands inside an isolation chamber they could not see. In March of 2022, in an undergraduate competition with 200+ other groups, we presented a poster in Texas A&M University's Student Research week and won second place in mathematics, engineering, technology, and computer

science. Unlike my previous research experiences, this project allowed me to participate in discussion and deliberation about the purpose of the pilot experiment with the post-doctoral researchers working with us. I learned a lot about important considerations for experiment design, as well as the difficulties of working with human subjects. I think this experience improved my understanding of how challenging it is to work with humans, and how performance can be quantified and analyzed.

Graduate Research: I am currently focused on researching the effects that pulsed, low frequency electromagnetic fields have on neural differentiation and growth, with an emphasis on proteomics. As I continue, I hope to investigate repairing brain tissue, aging, and the emergent property of consciousness. I firmly believe that furthering our knowledge on how we can manipulate neural cells and tissues will be a new frontier of medicine. I am currently developing a high-throughput protocol with the goal of aggregating and compartmentalizing the many chaotic procedures and parameters used in electromagnetic stimulation of neural cells towards a unifying theory.

D. Scholastic Performance

YEAR	COURSE TITLE	GRADE
2025	Mechanics of Biostructures 1	A
2025	Biomedical Modeling	A-
2025	Topics Biomedical Engineering	A
2022	Bioreactor Engineering	A
2022	Numerical Methods	B
2022	Project Management for Engineers	B
2022	Contemporary Moral issues	A
2022	Hydraulic Power	A
2022	Fundamentals in Bioseparations	B
2022	Engineering Design II	A
2021	Engineering Design I	A
2021	Engineering Properties of Biological Materials	B
2021	Design of Machines and Structures	B
2021	Fundamentals of Microbiology	B
2021	Technical Business Writing	A
2021	Linear Algebra	B
2021	History of Rock	A
2021	Research (Mechanical)	A
2021	Organic Chemistry Lab	F
2021	Measurement and Control	C
2021	Transport Procedures of Biological Systems	C
2021	Unit Operations	C
2020	Research (Mechanical)	A
2020	Principles of Electrical Engineering	B
2020	Mechanics of Materials	B
2020	Fluid Mechanics	A
2020	Bio & Ag. Engineering Fundamentals II	A
2020	Engineering Thermodynamics	B
2020	Introductory Biology II	B

YEAR	COURSE TITLE	GRADE
2020	Materials Science	A
2020	Differential Equations	C
2020	Introductory Biology I	A
2020	Bio & Ag. Engineering Fundamentals I	A
2019	Physics III Lab Electromagnetism	B
2019	Statics & Particle Dynamics	C
2019	Calculus III	C
2019	Organic Chemistry I	B
2019	Analysis of Bio & Ag. Eng. Problems	B
2019	Electromagnetics and Magnetism Physics	A
2019	State and Local Government	B
2019	Physics II Lab Mechanics	A
2019	Newtonian Mechanics	C
2019	Calculus II	C
2019	Fundamentals of Chemistry Lab	A
2019	Fundamentals of Chemistry II	C
2018	Calculus I	B
2018	Sociohorticulture	A
2018	Engineering Lab – Computation	B
2018	Fundamentals of Chemistry I Lab	B
2018	Fundamentals of Chemistry I	B

The first three courses listed are graduate coursework have been taken at the University of Central Florida that uses an A(+/-) system, and the remainder are courses taken at Texas A&M University using a Standard letter grade system with no + or – addendum.